

EX. NO: 12

WORKING WITH SNORT TOOL TO DEMONSTRATE INTRUSION DETECTION SYSTEM

AIM

To study and demonstrate the working of the Snort Network Intrusion Detection System (NIDS) using different operating modes.

ALGORITHM

Step 1: Start the system and install Snort.

Step 2: Configure the Snort environment and set the system PATH to the Snort installation directory.

Step 3: Run Snort in Sniffer Mode to capture and display network packets.

Step 4: Run Snort in Packet Logger Mode to store captured packets in a log directory.

Step 5: Configure the snort.conf file with the required network settings and rules.

Step 6: Run Snort in Network Intrusion Detection System (NIDS) Mode using the configuration file.

Step 7: Monitor the network traffic and detect suspicious packets based on predefined rules.

Step 8: Observe the generated alerts and log files.

Step 9: Verify the intrusion detection results.

Step 10: Stop Snort and terminate the program.

PROGRAM:

CODE / COMMANDS

1. Sniffer Mode

```
snort -v
```

Displays the TCP/IP packet headers.

```
snort -vd
```

Displays packet headers with application data.

```
snort -vde
```

Displays Ethernet, IP, TCP/UDP/ICMP headers and packet payload.

2. Packet Logger Mode

Create a log folder:

```
mkdir C:\log
```

Run Snort in packet logger mode:

```
snort -dev -l C:\log
```

Log packets from a specific network:

```
snort -dev -l C:\log -h 192.168.1.0/24
```

Binary logging:

```
snort -l C:\log -b
```

3. Network Intrusion Detection System (NIDS) Mode

```
snort -d -l C:\log -h 192.168.1.0/24 -c C:\Snort\etc\snort.conf
```

OR

```
snort -A console -q -c C:\Snort\etc\snort.conf -i 1
```

OUTPUT:

Running in packet dump mode

Commencing packet processing...

Packet Statistics -----

Received Packets : 30

Analyzed Packets : 30

Dropped Packets : 0